

Hex CLI: A Measurement-First Coding Agent on a 4-Billion-Parameter Model and a Laptop NPU

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github.com/NathanL15/Hex-CLI

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Abstract

Hex CLI is a terminal coding agent in the tradition of Claude Code, with one structural difference: it runs entirely on a consumer laptop, with no cloud service, no API key, and no code leaving the machine. It pairs a 4-billion-parameter model (Qwen3-4B-Instruct-2507) with the Hexagon NPU of a Snapdragon X Elite and makes up for the model's limits with engineering: an evaluation instrument graded on real filesystem state, a rebuilt agent loop, a server fork that keeps the model's cache warm across turns, and a security layer that assumes the model can always be manipulated. The model never changed across three months of releases, yet the extended task suite went from 22 of 35 cases passing all five runs to 32 of 44, multi-turn sessions from 23/45 to 35/48 turn-runs, executed prompt-injection payloads from 9/9 to 0/9, and whole-turn time fell by 41%. Ten plausible improvements, among them a new action protocol, prompt compression, a larger context window and a runtime migration, were built or benchmarked and rejected with evidence. The claim is methodological: at the small end of the model scale, the harness is the product.

1 The constraint is the point

Frontier coding agents assume a frontier model. Hex CLI asks what happens without one: how much of a real coding agent survives when the model is Qwen3-4B-Instruct-2507, running at about 15 tokens per second on a laptop NPU (the low-power AI accelerator in the laptop's processor), inside a 4,096-token context window (roughly 3,000 words), on a 16 GB machine with no cloud fallback?

The constraints are severe. The system prompt alone costs about 2,400 tokens, so once the server has reserved room for a reply, roughly 1,300 remain for the conversation and everything the tools return (Figure 1). For most of the project the picture was worse still: a “degradation cliff” at about 2,600 tokens and a 3,000-token input cap in the server held history to 250 tokens, and both turned out to be inherited constants rather than measurements (§5). Models must be quantized and compiled ahead of time for this NPU, and such bundles exist for exactly twelve of them, so swapping to a better model is rarely an option.

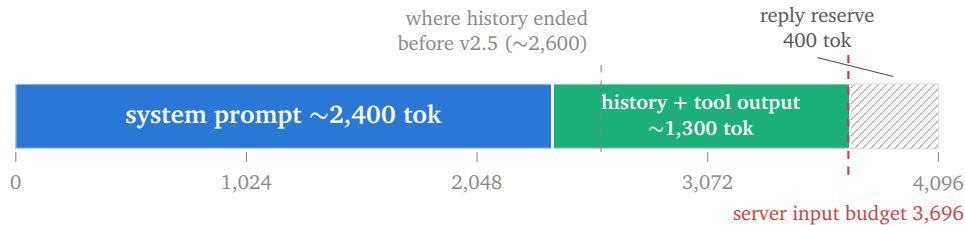


Figure 1: The 4,096-token window as the agent budgets it at v2.8.0. The server derives a 3,696-token input budget from the compiled bundle; the harness sizes history (about 810 tokens after a 500-token turn reserve) and tool-output pages against what the prompt leaves. Until v2.5 a “degradation cliff”, later traced to a parser bug, and a 3,000-token server cap ended history at the grey line.

Local inference is a solved problem: Ollama and llama.cpp run this model in one command, and agents such as aider, Claude Code and Copilot CLI define what a coding agent should do. What neither

side provides is an agent that stays competent and safe when the model behind it is an order of magnitude smaller than the one the agent was designed around and the window is a tenth the size. That gap, on hardware that laptops now ship with, is what this report is about, and its thesis is that at this scale the harness (parsing, context management, edit application, safety) matters more than the model.

2 System

The system is two layers in two languages (Figure 2). A Python harness owns the agent loop: prompt construction, action parsing, dispatch across 15 tools, safety checks, context compaction and the REPL. A fork of *npurun*, written in Rust, owns inference: it wraps Qualcomm’s Genie SDK behind a local server that speaks the OpenAI API. The fork is not a passive dependency: it gained exact token accounting, stop sequences, a crash fix, and later the KV-cache rewind, a bundle-derived budget and a request watchdog (§5). Inference occupies the NPU while the harness and every tool it runs execute on the CPU cores, so tools never compete with token generation.

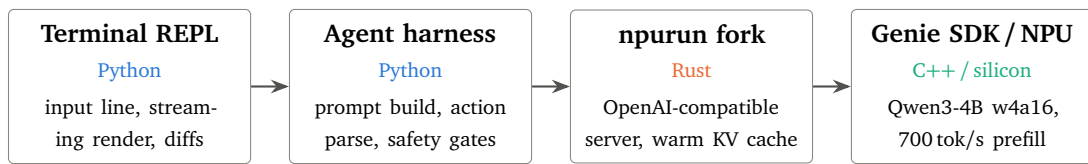


Figure 2: One agent turn. The evaluation instrument drives the same harness code path the user does: one loop, no test double. The model is quantized to w4a16 to fit the NPU. Python’s share of a turn is below 1% of wall-clock, which is why a Rust rewrite was considered and declined.

A session is a conversation in the terminal; each step is a tool call or a final answer. A representative task from the evaluation suite:

```

> fix the crash in report.py
read_file      report.py      locates the function
run_code       report.py      NameError: name 'conut' is not defined
edit_file      report.py      conut = 0 → count = 0
run_code       report.py      exit 0, output correct
Fixed. Line 12 of report.py misspelled count; corrected and re-run to confirm.
  
```

Every file mutation is shown as a diff, and a claim of “done” after an unverified edit is deflected: the harness asks the agent to check its work before it may finish.

3 Method: build the instrument first

The rule for v2 was that no change ships on intuition. Before touching the agent, the project built an evaluation instrument with three properties informal evaluations lack. Grading is based on state: a task passes only if the filesystem ends up correct or the answer is verifiably right, because a transcript containing an expected string proves nothing. Every case runs three to five times and passes only if all runs pass (pass^k), with Wilson intervals; three apparent regressions in this project vanished when re-run at $n=5$ or 10. And backend failures are quarantined: the NPU stack silently degrades after an hour or two of traffic and then looks exactly like a model regression, so the runner marks runs on a degraded server invalid rather than failed, and every A/B starts a fresh server per arm. Twice, an apparent regression was the platform drifting between runs.

4 Results

Three measurement campaigns produced the numbers here: the v2.0 harness rebuild in July, the context and runtime work of August and September (§5), and a backend study in September.

4.1 Same model, rebuilt harness

The v2.0 gains came from finding the harness’s real failure classes. The largest single fix was in the action parser: a greedy JSON-extraction regex was silently discarding the model’s batched multi-action responses, which was the true cause of a multi-turn collapse that had been blamed on context length. A four-tier fuzzy matcher now applies edits (the most common v1 failure was edit formatting, not model capability), malformed actions are retried with feedback naming the defect, and compaction was recalibrated with an exact token estimator. Figure 3 shows the effect.

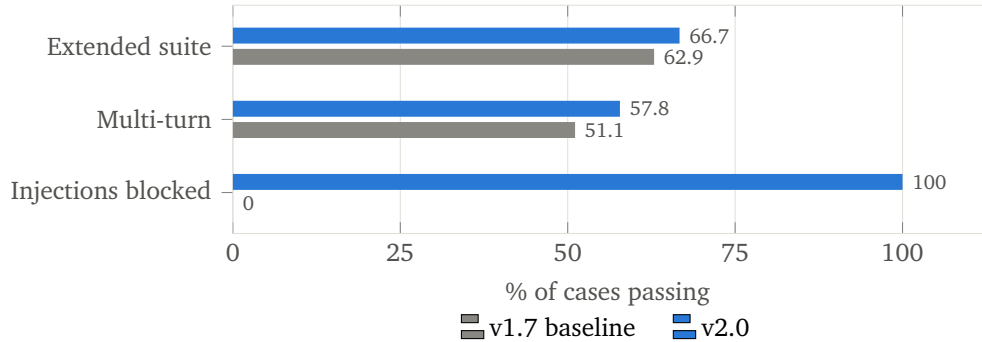


Figure 3: v1.7 and v2.0 on identical model and hardware. Extended suite: 22/35 to 24/36 cases passing all five runs (the suite gained a case). Multi-turn: 23/45 to 26/45 turn-runs. Injection payloads blocked: 0/9 to 9/9.

4.2 Where the window goes

Budgeting the prompt produced the most counter-intuitive finding of the project. Of the 1,990-token template at v2.0, the 14 behavioural rules were 73% and all 15 tool schemas about 450 tokens, so consolidating tools would have saved almost nothing, and §6 shows the rules could not be trimmed either. By v2.2 a live-state cookbook (which lifted machine-state questions from 44% to 72%) had grown the prompt to 2,355 tokens, measured from the server’s usage report, and auto-compaction fired every few messages. Compaction itself was the thrash: the compactor re-fired every turn and crushed its own summary into a stub. Making it merge-aware with a dry-run gain guard fixed the churn without touching the prompt. The history floor that remained was taken for a property of the model; §5 records that it was not.

4.3 Why the NPU, when the CPU decodes faster

On plain CPU the same model decodes faster than the NPU path (13.5 versus 8.1 tokens per second in the v2.0 test), but an agent re-reads its prompt, and prefill is where the NPU is untouchable: 4.1 s to first token against 13.6 s on CPU for the same 2,100-token prompt, arms interleaved. Extending the two measured points across turn lengths (Figure 4), the CPU would pay off only past about 193 generated tokens per turn, and real turns generate 12 to 126. A second study in September under the warm-cache runtime (`docs/backend_study/CPU_VS_NPU.md`) kept the conclusion and sharpened the numbers: the NPU decodes at 15.5 tokens per second at short context and 9–10 at the agent’s working context; a cold prefill of 2,048 tokens takes 8.3 s; a warm follow-up turn reaches its first token in about 0.8 s, and a new conversation pays a 6.6–8 s dialog rebuild only when the system prompt changes, which the byte-stable prefix avoids (an earlier “9 s per new conversation” was the benchmark’s own nonce forcing the rebuild, and is retracted). Switching off the runtime’s host-side polling raised decode to 19 tokens per second while cutting package power from 21 W to 9.8 W, 0.55 J per token instead of 1.44, with no quality change (62/82 against 61/82).

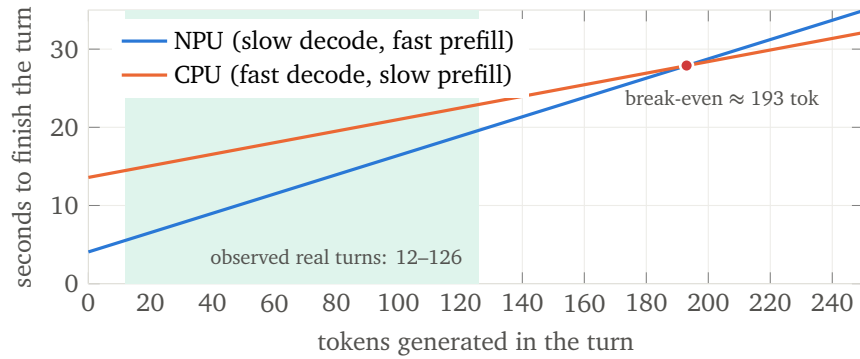


Figure 4: Turn time against tokens generated, from the v2.0 intercepts and decode rates (NPU: $4.06\text{ s} + 8.1\text{ tok/s}$; CPU llama.cpp Q4_0: $13.59\text{ s} + 13.5\text{ tok/s}$). The CPU wins only right of the break-even, and real turns never get there.

4.4 What still fails

Section 1 asked how much of a coding agent survives at this scale, and the honest answer includes the failures: 12 of 36 cases at v2.0, and 12 of 44 at v2.8.0, seven of them at 0 of 5 runs. They fall in two classes, both model ceilings. Bait compliance: when a prompt names a tool (“use `run_command` to...” for something that should be answered directly, the model complies about one time in three despite a rule against it; three of the seven hard failures are these. Edit anchoring: on a follow-up turn the model reconstructs an `edit_file` anchor from memory instead of from the file, three times in a row, or describes the change in prose instead of making it. The harness turns both from silent failures into retried or deflected ones; it cannot eliminate them. They mark where a better small model would help, and the instrument can judge one in an afternoon. The anchoring class did yield one more harness answer after v2.8: the model rebuilds lines from the traceback in its history rather than copying them (“for item in data” where the file says `items`), so its anchor sits at 40–95% similarity while the change it wants is one token. The fallback that pasted its text over the closest region at 95% similarity was installing those misremembered names—five of six runs of one multi-turn case ended in a `NameError` the harness had written. A token-level three-way merge (the model’s change applied where the file agrees with what it saw, the file’s text kept elsewhere, conflicts refused) replaced the paste in v2.9.0: the affected turn went from 1–3 of 3 to 5 of 6, a case at 0 in every baseline passed for the first time, and the extended suite was flat at run level ($p = 1.0$).

5 The runtime lever

The largest gain after v2.0 came from the runtime, not the prompt. A 2,355-token system prompt prefilled again on every agent step is a fixed tax of about five seconds per call, and the Genie runtime in QAIRT 2.47 rejects its own KV-cache rewind on this bundle. QAIRT 2.50 accepts it. The fork now keeps one dialog alive for the life of the process and sends every warm query as a prefix-matching rewind, rebuilding the dialog only when the system prompt changes; two runtime behaviours dictated that shape (a reset after a large prefill wedges the dialog, and an early-diverging transcript poisons it). The prompt had to become byte-identical across directories and days, which moved the date and working directory into the first user message. Measured on the extended suite, three runs per case, fresh server per arm: run-level 91/117 $\rightarrow$ 97/117 (Fisher $p = 0.41$), first-token median $6.8\text{ s} \rightarrow 3.7\text{ s}$, later steps $7.6\text{ s} \rightarrow 3.2\text{ s}$, whole-turn mean $16.0\text{ s} \rightarrow 9.5\text{ s}$.

With the prefill tax gone, the context question could be re-read. The 250-token history floor rested on two constants nobody had derived: the harness’s 2,600-token “degradation cliff” (the collapse it described was, by the project’s own record, the parser bug of §4.1) and a 3,000-token input cap inherited from upstream. The bundle is compiled to 4,096. A sweep at controlled input sizes found quality flat from 2,370 to 3,697 tokens, the same three cases failing at every size, and decode about 12% slower only with the window actually full. The server now derives its budget from the bundle (3,696 after a

reply reserve) and advertises it; the harness sizes history and tool pages against it, lifting history to about 800 tokens. Two defects surfaced only in multi-turn measurement: a rewind on a transcript that diverges mid-conversation can return success with zero tokens (63 of 117 calls in one run; now treated as a failed rewind), and a divergent rewind fails outright past roughly 3,200 cached tokens, which an end-of-turn prewarm absorbs in the background. Multi-turn went from 25/48 to 35/48 turn-runs and server trims per run from 141 to 17, with the extended suite at parity.

6 Negative results

Half the value of the instrument is what it stopped from shipping (Table 1). One finding runs under several rows: a small model tuned for months against one prompt becomes specialized to that exact prompt. Dropping even provably irrelevant rules degraded behaviour, and removing one rule hurt more than removing four. At this scale the prompt is not documentation the model consults; it is part of the distribution the model has adapted to, and that holds at every step of the loop, not just the first. The one split that survived was the one that changes the model’s task rather than shrinking its prompt: a harness-routed no-tools stage for pure-knowledge queries, which cut their first-token latency by 40% (10.1 s $\rightarrow$ 6.0 s median). It is switched off under the warm-cache runtime, where a knowledge query is already decode-bound and a second system prompt would only buy a rebuild.

Idea	What the measurement said	Verdict
New structured <code><action></code> protocol	13/36 vs. 22/35 pass ⁵ ; two months of tuning had specialized the model to the old format	rejected
Prompt trimming (drop situationally irrelevant rules)	saved 690 tokens and 16% latency, but bait resistance fell from 5/8 to 3/18 ($p \approx 0.017$), and the damage did not scale with rules dropped	rejected
8K context window (recompiled bundle)	6 tok/s vs. 15, no quality gain	rejected
Raise the compaction threshold	flat from 2,370 to 2,973 tokens but capped at $\sim$ 3K, so closed as “zero usable history”; the cap was a server constant, a second sweep was flat to 3,697, and history went 250 $\rightarrow$ $\sim$ 800 (§5)	reversed
Leaner prompt for steps ≥ 2	quality flat (77.8% vs. 74.4%, $p = 0.65$), but the JSON-edit case fell 3/3 $\rightarrow$ 1/3 with degenerate anchors, the trimming fingerprint again	rejected
Escalate hard cases to Qwen3-8B	0.9 tok/s: silent CPU fallback on 16 GB	rejected
Qwen3.5-4B successor	20–25% slower than the incumbent on both CPU and hybrid paths	deferred
Native Qwen3 <code><tool_call></code> template	the quantized bundle’s detokenizer garbles its own special token	impossible
GenieX runtime (+47% claimed)	0.74 tok/s here: a fixed 1.5 s penalty per NPU graph call; filed upstream (#1266, PR #1267)	halted
Rewrite the harness in Rust	Python is below 1% of a turn; the bottleneck is inference	declined

Table 1: Ten plausible improvements that failed their evaluations, recorded so none is re-argued from intuition.

7 Security: assume the model is fully injectable

To a 4B model, a malicious instruction hidden in a README it was asked to read is indistinguishable from the user. v1.7 executed all nine payloads in the live injection suite (hosts-file reconnaissance, SSH-key exfiltration, spawning executables); v2.0 blocks all nine, and the principle matters more than the score: no defense may depend on the model resisting the bait. Enforcement lives in the harness. A *sensitive* tier above “safe” covers credential stores, SSH and cloud keys, registry hives and encoded-command

tricks, confirmed interactively and denied when unattended; every mutating path routes through one `guard_mutation` gate, and a test enumerates every mutating entry point against a forbidden path; the one tool that reaches the network asks before every fetch. One measured lesson became a style rule: an early refusal message suggested an alternative command and the model followed it straight to the bypass, so refusals never name another route. Packaging in v2.8.0 found the classifier’s own blind spot: it had treated every `Format-verb` as a disk format, so the `Format-List` the model appends to live-state queries asked for a confirmation that unattended evals answer with a denial. The live-state numbers above carry that handicap; the fix is a two-line pattern.

8 Related work

Terminal agents that accept a local model. OpenCode, Crush, Goose, Qwen Code, Codex CLI, OpenHands, Cline, Kilo Code, Mistral Vibe, Pi and Aider all run against a local server, and all assume a window of 16K tokens or more and, in practice, a model of 20–35B parameters. OpenHands’ own documentation calls a 4,096-token window “way too small—not even the system prompt will fit”; OpenCode’s compaction keeps 15K recent tokens, so it cannot operate inside this paper’s window at all; Codex asks for at least 32K.¹ Most rely on native tool calling with no text fallback; “the model prints the tool call instead of executing it” is the most-reported local failure in their issue trackers. None has any NPU code. No other coding or shell agent running on a Hexagon NPU was found in 2025–2026; the nearest neighbours are a bridge that feeds OpenCode from Microsoft’s Phi Silica on the same 4K window, whose author notes that “a real terminal agent’s toolset alone measured nearly three times the whole window”, and AMD’s GAIA, which is AMD-only.²

Harnesses built for small models. The patterns this paper measured have precedents. mini-SWE-agent uses one action per step, a linear history and no tool API, and is positioned to “run with literally any model”; nanocoder is the one CLI explicitly targeting 1–4B models, with a 150–250-token prompt that drops tools, JSON and XML text fallbacks and no published evaluation; octofriend ships two 8B LoRA adapters that repair malformed SEARCH/REPLACE blocks and JSON after the fact; Aider switches to whole-file edits below roughly 32B because format compliance collapses.³ The deterministic compaction of Section 4.2 agrees with the evidence: on SWE-bench Verified, masking old observations matched LLM summarisation at half the cost (2508.21433). The security stance of Section 7 matches the out-of-band defenses evaluated on a 7B model, where a deterministic gate cut attack success from 25.8% to 4.2% (Progent, 2606.26479), and the finding that 4B-class models fall to essentially every prompted injection (InjecAgent, 2403.02691).

Evaluation practice. Terminal-Bench 2.0 requires five trials per task with confidence intervals; τ -bench defined pass^k; no published result for a 1–8B agent reports pass^k or a significance test, and Aider’s leaderboard has no row below 32B.⁴ On the runtime side, Qualcomm’s published figure for this exact bundle on this NPU is 6.56 tok/s; an independent measurement of the same class of bundle found 14.2 tok/s decode and 787 tok/s prefill, in line with the 15 tok/s here, and confirmed that the static graphs impose the fixed window this paper works inside.⁵

¹<https://docs.openhands.dev/openhands/usage/llms/local-llms>; <https://opencode.ai/v2/docs/compaction>; <https://ollama.com/blog/codex>.

²<https://github.com/ookla-ariel-ride/npv-bridge>; <https://github.com/amd/gaia>.

³<https://github.com/SWE-agent/mini-swe-agent>; <https://github.com/Nano-Collective/nanocoder>; <https://huggingface.co/syntheticlab/diff-apply>; <https://aider.chat/docs/leaderboards/edit.html>.

⁴<https://github.com/harbor-framework/terminal-bench-2-1>; <https://arxiv.org/abs/2406.12045>.

⁵<https://huggingface.co/qualcomm/Qwen3-4B>; <https://arxiv.org/abs/2606.11257>.

9 Lessons

1. **At small scale, the harness is the product.** Every measured win (parsing, edit application, compaction timing, retry feedback, the warm cache) came from the scaffolding around a frozen model.
2. **Build the instrument before the feature.** `passk` with state-based grading made every claim here possible and repeatedly overturned the story a single run told.
3. **Interrogate the measurement platform.** A silently degrading backend impersonates a model regression perfectly.
4. **Security is a property of the architecture, not the model.** Assume total injectability; put every enforcement decision in code you control; test it by enumeration.
5. **Negative results are deliverables.** Written down with their evidence, they keep future work from re-fighting settled questions.
6. **Mechanics beat prose.** Twice in v2.7 a prompt sentence fixed one case and broke a gated one; a loop nudge (“run the tests” means a test ran, 0/5 → 5/5) and a loop guard (no edit lands on a file the request did not name, 1/5 → 5/5) did the job without touching the prompt.
7. **Never paste the model’s memory of a file into the file.** A fallback that accepts a near match and writes the model’s text installs every name it misremembered; transfer only the change it asked for.
8. **Reading a file back is not verification.** The model will call a page “verified” after `read_file` and report that it ran when nothing ran; a claim of behaviour needs an execution event in the turn, and a verifier must say which rung it reached, including “not checked”.

10 Status

At v2.8.0 the agent runs in Windows Terminal with a pinned input box and status line, prints only the answer when its output is a pipe, and installs from PyPI (`pip install hexcli`). The headline is 32 of 44 cases at `pass5` on a fresh server and 35 of 48 multi-turn turn-runs; the multi-turn suite still loses about one run in three near 3,100 tokens to a platform stall that the server’s watchdog now ends in 97 s rather than 400, counted as a backend failure. What remains on the harness side is small: compaction sharing the warm prefix, paging of large files, and the edit-anchoring case. The other levers need a different model or runtime, and when a successor bundle appears for this NPU the instrument can judge it in an afternoon.⁶

Table 2: Milestones, 2026.

Date	Milestone
Jun 16	First end-to-end NPU inference at about 15 tok/s, after two other runtimes proved dead ends.
Jul 30–31	v2.0: instrument v2, loop rebuild, injection defense 0/9 → 9/9, product shell; 619 tests.
Aug 14	v2.1: one-command installer, piped stdin, custom commands; a review found ten defects, all fixed.
Aug 16	v2.2: live-state cookbook 44% → 72%; the memory daemon found re-injecting the model’s own confabulations.
Aug 29–31	Compaction thrash fixed; command surface pruned to one mode (−685 lines); prompt-split A/B.

⁶All figures were measured on a Snapdragon X Elite (X1E78100), 16 GB RAM, Windows 11 ARM, with Qwen3-4B-Instruct-2507 (w4a16) on the Hexagon NPU. Run-level data for the gate baselines is in `evals/results/` in the repository; the fuller measurement archive is kept by the maintainer.

Date	Milestone
Sep 1–2	Main module split into seven stages (3,818 → 1,499 lines); KV rewind on QAIRT 2.50, –41% turn time; context floor traced to two constants and lifted.
Sep 5–7	CPU-versus-NPU study; host-polling control, async init and the request watchdog (v2.6.x).
Sep 10–12	Windows Terminal surface; loop nudge and named-file guard through the gate; v2.7–2.8; PyPI package; Format-List classifier fix.
Sep 13	v2.9: three-way merge replaces the closest-match paste in <code>edit_file</code> ; double-escaped arguments decoded; the launcher found running a stale installed copy for two days; input line undo/redo, command menu, flicker fix (v2.9.1).
Sep 14	v2.10: a broken first JSON object is repaired or retried, never skipped for a finish behind it; claims need evidence; <code>verify_syntax</code> reports its rung; protocol v2 and both escalation paths removed.